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Week4&5 - Report

Problem definition: what you are predicting/estimating/explaining and what the inputs/outputs are.

I am trying to predict occupational employment in 2024. The inputs I used in this version of the project are minimal (to not overload the script) but they include Mean, Median, Entry, and Experienced wages lagged by one year. I would like to restructure the data in the future so that previous years employment can be predictors and I would like to add technology skills and work context from O*Net Online.

Data description (initial): what data you plan to use, key variables/features, and expected sample size.

I plan to use Occupational Employment and Wage Statistics from the Bureau of Labor Statistics. I will be using Florida statewide data for every occupation and using the industry totals. I will be collecting the employment, hourly and annual mean wages, and the hourly and annual median wages. I expect to have the entire data set available to me and to use it.

Preprocessing plan: cleaning steps, missing data handling, encoding, normalization/scaling, train/test split strategy, etc.

Not much preprocessing has been done at this point but I plan to add more features in the future that will need more processing. As of now, the preprocessing done includes changing numeric features to numeric data types, removing NAs, and lagging wages to the previous year. Modeling/analysis plan: algorithms/methods you will use and why they fit the problem. I plan to use a few models. While I expect Random Forest or XGBoost to be the most successful because they capture non linear relationships and based on the literature review, they are able to account for situational outliers, I also used

Linear Regression
Ridge and Lasso Regression

Evaluation plan: metrics (e.g., accuracy, RMSE, AUC), baselines, and how you will validate results.

I use the cross validation method to validate the training data set

Metrics used to validate results include

Root Mean Squared Error (RMSE)

Mean Absolute Error (MAE)

R-Squared (RSQ)

Provide the dataset source (paper link, repository link, or dataset citation).

<https://www.floridajobs.org/workforce-statistics/data-center/statistical-programs/occupational-employment-statistics-and-wages>

Option B: test your code on your chosen dataset and report preliminary results (even if the paper you chose does not provide a direct benchmark).

Your report must include:

what you implemented (brief description of the pipeline),

After reading in the data, I lagged the wage features to the previous year. Then I split the data so that the training data set includes 2015-2023 data and the testing data set includes only 2024 data. Then I used cross validation to resample the training data. Next, I

created the models and tuned their parameters and then fit all the models. Next, I evaluate the models, determine the best one, and use it on the testing data.

what you tested (which dataset, which subset, which variables),

I used the dataset combining all years (2015-2024) and lagging the wage variables. The outcome variable is Employment and the predictors are Mean, Median, Experienced, and Entry Wages lagged. I would like to add to the features, but I need to find the data for these features (Educational requirements (BLS), Technology Skills (O*Net), and Work Context(O*Net)) what worked and what did not (bugs, limitations, unexpected issues),

On the whole, the script worked. The only unexpected issues were that it took a long time to run and there were some packages I wanted to use that I needed to edit my R Profile to be able to install. Random Forest took too long to set up so I had to eliminate it but I think XGBoost will be a good replacement.

preliminary outputs (tables/figures/metrics).

Based on the preliminary outputs, I selected XGBoost to use as my model.

I also will need to add several features because the R-Squared (0.0139) is low.